Goal

In this lab, I used colour histograms to do object detection for some test images. I used two colour methods (HSI and RGB) which were both tuned to produce “optimal” results

Algorithm

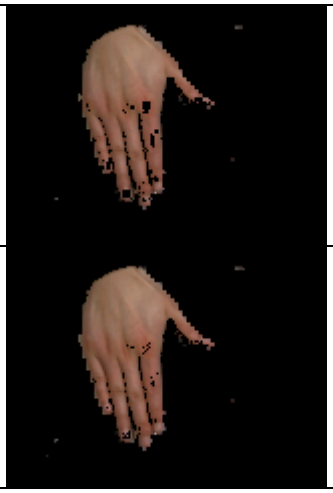
The solution first begins with training to create a normalised histogram. For both RGB and HSI, I looped over every pixel and incremented a counter in a matrix which has the size corresponding the space of the colour type.

Next, this matrix is normalised using val – min / (max – min) to scale values from 0 to 1.

Finally, our test images are analysed pixel by pixel and their RGB / HSI value are used to find their normalise value within the matrix. If this value is higher than the threshold then that pixel is kept.

Results

Below are the results for HSI and RGB. The threshold was used to find the best results by eye however the training data is also included in my submission for reference.

HSI Thresh = 0.0001			
RGB Thresh = 0.001			